Simlaw Seeds®

Superior & Reliable

TOMATO PLANTING GUIDE

2020

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PREFACE

Simlaw seeds Company ltd, a subsidiary of Kenya seed company, was founded in 1929 making it one of the oldest seed companies in East Africa region and renowned for supplying 'superior and reliable seeds' to farmers. Over the years of operation, Simlaw seeds has managed to maintain it's position as a leader by closely collaborating with it's partners across the world, being in constant touch with all its stake holders and most importantly, investing in research and development.

It is our pleasure to be associated with the farming success in the East Africa region in line with our vision and mission of availing superior and affordable certified horticultural seeds and other inputs to improve agricultural productivity and livelihood of stakeholders. Key amongest the stakeholders is ' the farmer.'

This planting guide provides a summary of our cabbage varieties available from Simlaw Seeds co. and it's outlets.

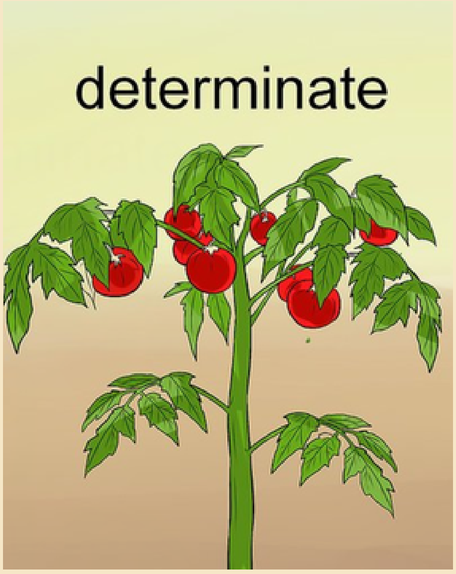
INTRODUCTION

Tomato belongs to the family Solanaceae. It is one of the most important vegetables grown in Kenya. Its fruits are used fresh in salads or cooked as a vegetable, processed into tomato paste (puree), tomato sauce, ketchup, juice and can also be sun dried. Tomato is rich in vitamins A and C and is gaining importance because of lycopene, a food component known to reduce the incidence of prostate cancer.

CLIMATIC AND SOIL CONDITIONS

Tomatoes are fairly adaptable to different agro-ecological zones, it however prefers warm conditions with optimum temperatures ranging between 15°C to 25°C. Though high humidity and temperatures reduce fruit set and yields, low temperatures can delay color formation and ripening. Temperatures above 30°C inhibit fruit set, development and flavor. Tomato thrives best in low-medium rainfall with supplementary irrigation during periods of drought. Wet conditions increase disease incidence and affect fruit ripening. Tomatoes can do well in a wide range of soils, but prefers those that are high in organic matter, well-drained and a pH range of 5 - 7.5. Soil analysis should be done before fertilizer application.

CURTIVAR SELECTION



The farmers choice of cultivars is based on fruit quality, adaptability to environment and reliability, tolerance to diseases and pests, fruit firmness, plant growth habit, availability and suitability to the growing conditions, market preference, and the planting time. The choice of the best variety to plant is critical for good yields. Because there are many varieties in Kenya, you need to choose one that will meet your needs and are suited to your climate. For both processing and fresh market tomatoes can either be determinate or indeterminate. A determinate will set fruit at the terminal bud and growth stops. The harvest time of determinate is short normally sometimes only a week to ten days. An indeterminate does not set fruit at the terminal bud. The growing points produces leaves and more stems. Harvesting time may last for months and the vines will travel extensively over the ground or staking can also be done to get healthy and clean fruit. Semi-determinates produce stems that end with a flower cluster. These varieties grow slightly taller than the determinates.

VARIETIES AVAILABLE IN SIMLAW SEEDS

VARIETY	OPEN FIELD/ GREENHOUSE	CURTIVAR TYPE	HYBRID/OPV
PROSTAR F1	GREENHOUSE	INDETERMINATE	HYBRID
NYOTA F1	OPEN FIELD	DETERMINATE	HYBRID
MONICA F1	OPEN FIELD	DETEMINATE	HYBRID
LIBRA F1	OPEN FIELD	SEMI-DETERMINATE	HYBRID
CROWN F1	OPEN FIELD	DETERMINATE	HYBRID
RIO GRANDE	OPEN FIELD	DETERMINATE	OPV
CAL J	OPEN FIELD	DETERMINATE	OPV
M82	OPEN FIELD	DETERMINATE	OPV
MONEMAKER	OPEN FIELD	INDETERMINATE	OPV

RESISTANCE TO DISEASES

A Disease control is one of the major factors affecting tomato quality and production. It is wise to avoid varieties that offer no tolerance to serious disease prevalence in your area. Tolerance is usually indicated by whether the variety is a hybrid or an OPV.

HYBRID OR INBREED LINES

Hybrids are plants resulting from the crossing of two or more different biotypes of the same or different species. Seeds of OPV can be harvested and saved for future planting while those for hybrids cannot. Hybrids often produce higher yields, uniform fruits or any other preferred quality attribute. Hybrids are usually more expensive than OPV.

CROP MANAGEMENT

Avoid planting tomatoes in a field planted previously with tomatoes or plants of the same family like pepper, eggplant etc. This is because they share similar insect and disease problems.

Other crops not recommended include: potato, okra, tobacco, onion, garlic, melon, cucumber and squash. Crops recommended include – green beans, cabbage, turnip, cassava, maize, sorghum, groundnut and broccoli.

FIELD PREPARATION

Compacted soil should be thoroughly loosened by deep ploughing to enable the root system to spread to a depth of 40 – 60cm. Field leveling and instillation of drainage tiles should be done for sites with uneven topography, ditches and waterlogged sites. The field should be near a reliable water source as tomatoes require frequent watering. Low soil pH can be corrected by addition of Lime and high soil pH or Sodic soils remediation is done by application of Gypsum. Soil borne Pathogens and Nematodes can be controlled by soil solarization after a deep plough. False sowing technique can be used to reduce the number of weed seeds present in the soil. This is done by watering the field after full preparation. The weeds are then eliminated by hoeing or by chemical weed killer.

SEED RATE

Tomato is usually raised in the nursery which can be purchased as certified seeds from our agents and stockiest. These seeds are usually treated with spectrum fungicides to control damping off. For a acre, you require about 50gms of seed which has approximately 15,000 seeds of hybrid varieties or 100g of seeds for OPV varieties which has approximately 30,000 seeds.

SEED SOWING

Tomato seeds are either sown into a nursery or sown on plastic seeding trays. **1: Seed Tray Method**, In this method, the seed tray holes are filled with a media that drains well such as commercial potting soil, cocopeat, peat, moss or a mixture of sand and compost. Sow 2 seeds per hole and thin seedlings 2 - 3 days after the first true leaves. This method makes seeds healthier and more vigorous and suffers no root damage when transplanted. **2: Nursery / Seedbed method,** the nursery should be sited in a well-drained area preferably not previously planted with a Solanaceous crop. The seedbed or nursery site should be in a place protected from direct sunlight by erecting a shade structure.

Seeds should be sown in rows 6 cm apart at a rate of 750 - 900 seeds / m2. The seedbed surface is spread with a thin layer of compost or straw mulch. Water the seedlings regularly so that the soil is moist but avoid water logging. Thin the seedlings 2-3 days after the first true leaves appear. A week before transplanting, reduce the watering frequency to harden the seedlings. Transplanting should take place one month after sowing (4-6 leaves stage) the seedlings are uprooted with a ball of soil when they attain from 4 to 6 leaves. This should be done on a cloudy day or late in the afternoon to reduce transplanting shock, before lifting seedlings from the bed, thoroughly water to avoid excessive damage to the roots.

TRANSPLANTING

Harden your seedlings before to transplanting by reducing water application and directly sunlight exposure them to 6-9 days before. A good seedling ready for transplanting is in the fourth or sixth leaves stage (about 4 weeks old) and is vigorous and stocky. Thoroughly water the seedlings about 12 hours before transplanting to the field. This facilitates easy uprooting of seedlings. Transplant the seedlings in the late afternoon or early morning to minimize transplant shock. Insert the seedlings in a hole in such a way that the first real leaves are 5 - 10 cm above the surface. Plants can be planted on raised beds or flat land at a spacing of 0.6 - 1.2 m between rows and 0.45 - 0.9 m between plants. Press the soil around the root and water around the base of the plant. Irrigate the field moderately as soon as possible after transplanting.

MANURE AND FERTILIZER APPLICATION

In soils poor in organic matter use 10tons per acre (two handfuls per planting hole) of manure and mix thoroughly with the soil. Triple superphosphate should be applied at 50 kg per acre (which is equivalent to 10g or one teaspoonful per planting hole) and mixed well with soil. Top dress with CAN at 50 kg per acre (5gms or 1/2 teaspoonful per plant), when plants are 25 cm high and 50 kg per acre (10g or one teaspoonful per plant) after 4 weeks. Inadequate nutrient levels may result in deficiency disorders that can be observed as patterns of leaf discoloration or fruit abnormality. The major nutritional disorder in tomato is blossom end rot caused by calcium deficiency. Deficiency disorders can be corrected by application of foliar fertilizers.

CULTIVATION PRACTICES

Mulching: this conserves moisture and keeps down the soil temperature it is especially important for the determinate varieties to maintain the fruit quality.

Weeding: should be done regularly to reduce nutrient competition and to destroy alternate hosts of pests and diseases.

Training/Staking :Training of plants has a number of benefits, including improved spraying to control foliar diseases and pests, less sunburn, better air circulation around the plant and less fruit rotting. Cultivars that have an indeterminate growth habit require training on trellises or suspended on twine from overhead wires. The crop grown for the fresh market should also always be trellised.

Pruning: This is done for indeterminate varieties. The practice influences the flowering and fruiting of the tomato plant, leave one or 2 main stems and pinch out the laterals branches as they appear in the leaf axils. When 6-8 trusses are formed pinch out the growing tip to encourage growth of good size marketable fruits. Remove leaves close to the ground to help prevent the entry of blight.

*farm managers who smoke should wash their hands before handling plants as they may otherwise spread tobacco virus disease. It has been reported that pruning reduces early and total yield, detrimentally affects quality and increases the incidence of viral diseases and other disorders. This practice is not recommended for tomatoes grown in the open field.

CROP PROTECTION

INSECT PESTS

Crop rotation: A three-to four year-rotation program with non-related crops is recommended to reduce build-up of pests and diseases.

Irrigation: In dry weather regular watering is essential. The ground should be given a good soaking 2-3 times each week. Irrigation for tomatoes should be regularly especially during critical periods like flower setting and growth of the fruits. Watering is reduced at the end of crop maturity. Regular watering reduces blossom end rot, ensures uniform fruit development, prevents fruit splitting, reduces the risk of sun scorch, enhances fruit growth and increases the size and number of fruits. Excess water increases water logging which leads to Magnesium, Phosphorous and Nitrogen deficiencies. Drought during fruiting period is the most critical for the final yield. In very hot weather, sprinkler irrigation should be done early in the morning to avoid heat shock, burned leaves, non-setting flowers and setting up of a microclimate favorable for diseases. Irrigation can only be performed by the end of the day so long as plants dry up by nightfall to avoid late blight disease. In practical terms, apply 3-4mm per day at planting, 6mm per day at first.

Cutworm: The major soil pests attacking tomato seedlings are cut worms, Agrotis spp and chafer grubs, Melolontha spp. Cut worm larvae are grey to black-caterpillars approximately 24 mm long often found hidden in the soil near the seedlings. They hide in the soil feeding on the underground parts of the plant during the day and come to the surface to feed on the aerial parts of the plant. Soil pest infestations are sporadic and are more common in weedy spots, fields with high organic matter and poor drainage.

Damage:

They cut the seedlings stem at the soil line, and eat holes into roots. The injured plant wilt, wither and die. Young caterpillars feed on the leaves leaving perforations on the leaves. The pests feed on the plants at the base causing serious damage to stems. Stalks of plants may be cut. Soil pest infestations are sporadic and often associated with sections of the field that are weedy, have high amounts of organic residue, or poor drainage. On digging up the soil, the pests are found in the roots region.

- **Cultural control:** Fields need to be prepared and weeds eliminated at least two weeks prior to planting to reduce soil pest damage. Early detection and manual removal of observed soil pest. They can be found in the evening or scratch soil near plants to uncover them. Weed the field early also tilling and ploughing soil is necessary to kill soil pests. In fields with high infestations, put a 3 inch cardboard collar around young seedlings and push it 1 inch into the soil.
- **Bio-pesticides:** Bacillus thuringiensis (BT).

- **Chemical control:** 1. 5% Malathion dust around the plant after transplanting preferably in the afternoon. 2. Dipterex (Dylox) Trichorphon 5% dust similarly at 2 kg/ha and add baits e.g. Bran mixed with sugar. 3. Spray with pyrethroid insecticides at transplanting.

READ THE LABEL ON THE PESTICIDE CONTAINER CAREFULLY AND USE THE MANUFACTURER'S RECOMMENDED RATE.

African bollworm: The caterpillars of the African bollworm are major pests of tomato. The pest is extremely polyphagous on a wide range of crops including (cotton, maize, sorghum, sunflower and beans).

The caterpillars vary from dark green, reddish, brown, whitish and orange in color and have a characteristic undulating white band on each side of the body. They grow up to about 40 mm in length.

Damage:
The caterpillars cause damage to the floral buds, flowers and fruits. They bore into the fruit often with the hide part of the body exposed outside. They produce copious amounts of pellet-like grass. One caterpillar can cause damage to several plant parts such as flowers, flower buds and fruits which are characterized by presence of one or two rounded holes. The caterpillars bore into fruit and feed on the inner part of the fruit releasing plenty of excreta (frass) which is noticeable on damaged fruits

Control:

• Cultural control. Scout the field for bollworm caterpillars to initiate control before actual damage is incurred. Practice deep cultivation to destroy pupae in the soil. Avoid planting susceptible crops in succession.

• Natural enemies. Parasitoids such as Trichogramma, Trichogrammatoidea and predators (syrphid flies, ladybird beetles).

• Bio-pesticides. Use of microbial biocontrol agents is a potentially important compound of IPM for the African bollworm. The most commercially important include, strains of Bacillus thuringiensis (B.t.), nuclear polyhedrosis (NPV) Metarhizium anisopliae (M.a) neem products and insect growth regulators such as Match.

• Chemical control should target young caterpillars before they can enter the fruit. The African bollworm is one pest that develops resistance to same chemicals over a short time. Registered effective chemicals products available against the African bollworm include: Deltamethrin, Carbaryl, Bacillus thuringiensis, Spinosad, Bifenthrin, Indoxacarb, Pyrethrin, Methomyl and Novaluron among others. Farmers MUST adhere to the recommended tank dose and observe the Pre-Harvest Intervals as advised on the product label

*Chemical spraying should be done as soon as young caterpillars are noticed on the crop since control becomes increasingly difficult with older caterpillars.

Whiteflies: are small mealy insects of about 1.5mm long with two pairs of white wings.

Damage:

These small white scale like insects occur on the underside of leaves. The adults fly away from the leaf once the foliage is disturbed. The nymphs suck plant sap from the underside of the leaf.Their sap sucking activity may result in wilting and leaf malformation. . After sucking the plant sap, the insect excretes a sticky sugary substance known as honey dew, which ants feed on. Sooty mould quickly establishes in the honey dew and may spread all over the leaves. Crop plants are weakened by the feeding of large numbers of whiteflies, whilst the sooty mould impairs leaf function. They also cause leaf distortion and stunting if the attack is very early. This can be a serious problem especially in hot areas. The insect is also a vector of cassava mosaic virus, cotton leaf curl, tobacco leaf curl and sweet potato virus B.

Control:

•Natural enemies. Encarsia formosa is a tiny parasitic wasp of B. tabaci, in which case the parasitized larvae become transparent to brown in color. Another tiny wasp, Eretomocerus spp. parasitizes B. tabaci.

•Bio-pesticides. Neem products

•Chemical control. This pest is easily controlled using pesticides. It is important to completely cover the underside of the leaves with the pesticide being applied. High concentration of these chemicals may injure the crop.

Red spider mite: pose a major problem on tomato and other members of the Solanaceae family (eggplant, Chillies, Capsicums and Irish potato). The pest is extremely polyphagous attacking many cultivated and wild plants. Because of their great reproductive capacity, they are able to destroy plants within a short period of time. When left uncontrolled the farmer can lose his production within a week. Infestation often starts on the outside (border rows) of a plot. therefore, other adjacent (tomato) crops, wild plants and weeds can serve as a source of infestation. The mites can also be spread passively by irrigation water, dust storms, clothing and implements. The mites can also be spread by wind All the different stages of insect development are usually found together on the leaves at the same time. The pest develops very rapidly in warm, dry weather.

Damage:

The leaves are injured as a result of the mites sucking out valuable substances from the underside of leaves causing speckling and tarnishing and eventual leaf fall. Under severe attacks they will cause stunted growth and reduce yields. The problem is more acute during dry weather spells. The mites and its webbing, just visible to the eye, can be seen on the underside of the leaf. Spider mites may also cause spots on the fruits. Because of their small size (0.3-0.5mm), an infestation is often only noticed after the leaves have been discolored.

Control:

• Cultural practices. Regular scouting of the crop to determine the presence of the pest and the level of infestation in an early stage is a substantial element of IPM (Integrated Pest management). Burning of infested plants can be successful during the early stages of infestation when the mites concentrate on a few plants. The separation of infected crops and newly planted crops or nursery areas and the burning or removal of infected crop residues and weeds, also helps to minimize the problem.

Natural enemies such as predatory mites are effective in the control of Spider Mites. e.g. Phytoseilus persimilis has been very effective when used in the green house.
Inter-planting tomatoes with garlic or onion, practicing field hygiene, use of resistant cultivars and planting host plants of predatory mites such as pigeon peas do help bring spider mites on check. Heavy rain or irrigation can reduce their numbers

• Botanical pesticides. Botanicals such as Neem (Azadirachta indica) and Tephrosia sp. are currently evaluated in Kenya for their effectiveness in the control of red spider mite

• Chemical control. Curative and preventive treatments especially during the vegetative phase are advisable. Effective insecticides include Abamectin, Amitraz, Dicofol, Clofentezine, Bifenthrin, Tetradifon and Azadirachtin among others. Care should be taken when considering chemical control. Thorough understanding of the different available chemical formulas and their cost-effectiveness is required.Some of the available systemic pesticides have shown to increase red spider mite reproduction. In addition, red spider mite species rapidly develop resistance against the most common used pesticides and acaricides. It is therefore recommended to rotate acaricides with different chemical compositions. Spraying should be done weekly and at an early stage of infestation to be effective.

Leaf hoppers: Leaf hoppers (Empoasca spp) are green to yellowish green wedge shaped insects which jump off from foliage when disturbed.

Damage:

They suck the juices from the plant and excrete excess fluids in the form of honeydew. As with aphids, this sugary substance may support the growth of sooty molds. Leaf Hoppers spread Curly Top Virus. Young plants are severely stunted and die, while older plants turn yellow, leaves roll upward, fruit production ceases, and plants slowly die. This is a minor pest in most cases but in dry weather numerous hoppers feed on leaves, leaving a mosaic of white specks and stunting growth. These are possible vectors of virus diseases.

Control:

• Cultural control. Remove any affected plants and destroy them. Sticky traps can be made using honey, molasses or another sticky substance on a piece of stiff cardboard. Wave the sticky traps over the plants and the insects will stick.

- Natural enemies. Predators e.g. lacewings or praying mantis.
- Bio-pesticides. Spray plants with soap and garlic spray.
- Chemical control. Use effective chemicals in the market like Karate

Leaf miners: The adults are small black and yellow flies about 2mm long. They lay eggs which
hatch into small larvae that feed by mining between the upper and lower epidermis of the leaves making a tunnel as they move along. Occasionally the larvae can be within the leaf mine as it feeds.

Damage:

The act of laying eggs and feeding on leaves can kill seedlings and in older plants allows the fungal diseases to enter the leaves. Damage by "mining" causes whitish blotches inside the leaves, kills the leaves eventually making them fall prematurely. This reduces plant yield and exposes the fruits to sunburn.

Non-chemical control:

- Use of parasitic wasps such as Diglyphus ssp has proved effective. Use of yellow sticky traps or yellow basins filled with water attracts the adult leaf miner. These are later killed.Destruction of hosts such as old crop debris as well as having a rotation with non-host crops can help reduce leafminer populations in the crop.

Chemical Control-Leaf miners have an ability to develop resistance against pesticides very fast. Regular rotation of pesticides is therefore advised.
Effective insecticides include Abamectin, and Spinosad among others. Farmers MUST adhere to the recommended tank dose and observe the postharvest intervals PHI) as advised on the product label.

Tuta absoluta: Tuta absoluta is a devastating leaf miner on tomato crops. The pest can cause up to 50-100% yield reduction on tomato crops and its presence may also limit the export of the produce. It reproduces rapidly with a life cycle of 24-38 days, depending on the temperature, the minimum being 9˚c. The caterpillars are yellowish when newly hatched; later turn yellow green with a black band behind the head and the fully grown ones have a pinkish color on their back. Caterpillars mine inside the leaf, stem or fruit but exit to pupate. They can temporarily be temporarily found outside the leaf mines or fruits. Pupae are light brown and are found in the soil, on the leaf surfaces or in curled leaves or mines. Adults which are grey-brown are active at night and hide between leaves during the day.

Signs and symptoms:

Eggs are normally deposited on the underside of leaves

• Only green fruit is attacked by the larvae.

• Caterpillars prefer leaves and stems, may occur on fruit crowns or inside the fruit itself.

• Serious infection leads to the leaves dying off completely.

• Mining to the plant causes malformation and damage to fruit paves way for fungal infections, leading to fruit rot before or after harvesting.

• Rot due to secondary infective agents.

Management:

• Use of Bacillus thuringiensis have shown efficacy in controlling outbreaks as well as compounds of spinosad and imidacloprid.

• Use of sex pheromone traps is highly effective on the males, thus reducing the populations due to reduced fertilization of the females. Pheromone lures can be used for monitoring and mass trapping.

Aphids: Aphids green peach aphid suck plant sap, which can reduce plant growth; they also secrete honeydew, on which sooty moulds growth. Sooty mould on fruits reduces their market value. They transmit virus diseases during feeding such as the cucumber mosaic virus. In Kenya, aphids are occasionally found on tomatoes, but they are not considered an important pest in the farm of tomatoes.

Control :

• Conserve natural enemies. Aphids are usually kept under control by a wide range of natural enemies. In particular, avoid use of wide spectrum pesticides since they kill natural enemies.

• Use of sticky traps yellow in color

• Use reflective mulch. Reflective aluminum mulches deter aphids from landing on plants. The effect is lost once plants are large enough to cover the mulch.

Thrips: Thrips may also be a problem in tomatoes in Kenya. Thrips are small (about 1 to 2 mm long). They usually feed on the lower surface of leaves puncturing them and suck the exuding sap. They also attack buds, flowers and fruits. Attacked leaves have a silvery sheen and show small black spots (thrips excrements). Under heavy infestation attacked buds, and flowers usually fall off. Attacked fruits show speckling and small necrotic patches on the surface affecting fruit quality. Fruits may become deformed. Thrips feed on tomatoes at all stages, but their feeding on seedlings is particularly damaging. Heavy infestation can reduce stands of young seedlings in hot weather. Thrips of the genus Thrips and Frankliniella are vectors of viruses such as the Tomato Spotted Wilt Virus (the most economically important virus in tomato production) and the Tomato Chlorotic Spot Virus.

Control:

• Conserve natural enemies. Predatory mites (eg. Amblyseius sp.), anthocorid bugs (e.g. Orius spp.), and other predators such as ladybird beetles, lacewings and spiders, and the fungus Entomophthora are important in natural control of thrips.

• Monitor the crop regularly. Check plants daily in the nursery, and crop borders in the field. Be particularly vigilant at flowering. Pay careful attention to flowers and flower buds.

• Destroy thrips pupae in the soil. This helps reducing subsequent thrips populations. Plough and harrow before transplanting to expose pupae in the soil from previously infested crops to natural enemies and desiccation. Soil solarisation and flood irrigation (flooding previously infested fields prior to planting/transplanting) destroy a large proportion of thrips pupae present in the soil.

• If necessary spray bio pesticides. Neem and some other plant extracts are reported to control thrips. Spinosad, a bacterial derivative is effective in thrips control. However, timing of biopesticide application is important Thrips are difficult to control with insecticides due to their secretive habits

(eggs are laid in plant tissue, the larvae and adult shelter in the flowers and larvae pupate in the soil). Spraying early in the morning or in the evening and mixing the spray with a sugar solution (which attracts the thrips out o

the flowers) are reported to increase efficacy of sprays.

TOMATO DISEASES

Bacterial canker

This is caused by Clavibacter michiganensis subsp. Michiganensis(CMM). The bacterium can be introduced into fields on contaminated seed or on infected transplants. It may also survive in soil on infested plant material for at least one year. Symptoms do not appear until the disease is well established in the field. Splashing water and cultural practices such as pruning and tying (wounding) have led to epidemics. Like many other bacteria, canker development is favored by warm, wet conditions, especially when plants are wounded and inoculum is present. The disease can reduce yields by as much as 90%.

Symptoms

Bacterial canker is characterized by the symptomatic wilt of the whole plant.

Symptoms in young seedlings are difficult to see, but often start as dark water soaked

areas on leaves and stems. Systemic infections often start as wilting followed by

discoloration of the vascular tissue. Stems may split vertically with dark brown

necrotic cankers developing under certain conditions. The disease progresses as

follows:

Lower leaves wilt and turn downwards. The whole leaf dries; curls upwards turns brown, wither but still remain attached to the stems. Stems may split open and the pith is often found to be discolored. Fruit symptoms are lesions with raised brown centers surrounded with a opaque halo. The most diagnostic feature of bacterial canker is the formation of fruit spots. These spots may be confused with those caused by bacterial speck or spot. However, fruit lesions caused by bacterial canker are bordered by a distinct white halo. These white halos may disappear as the fruit
ripens.

Bacterial Wilt: Bacterial wilt caused by Pseudomonas solanacearum is one of the most economic diseases of tomato. This disease causes wilt of tomatoes as well as potatoes. It is mainly seed borne in tomatoes.

Symptoms: Bacterial wilt infects the whole plant. Symptoms first appear on the youngest **l**eaves and a rapid wilt of the whole plant occurs. Adventitious roots appear on infected stem. The vascular bundles will be yellowish brown in the initial stages of the disease. The pith of a completely wilted plant is brown. Bacterial wilt is distinguished from other wilts by suspending a part of infected stem in water whereby a slimy stream of bacterial cells from the xylem will be observed.

Control:

Practice rigid crop rotation (long term), tomatoes should not follow Solanaceous crops like potatoes, chillies and others i.e. practice crop rotation with a non-susceptible crop.

Produce transplants in pathogen free soil. Remove and burn infected plants as soon as possible to check the spread of the diseases. Plant resistant tomato cultivars-

Bacterial spot: This disease is caused by a bacterium Xanthomonas campestris pv. vesicatoria (Didge) Dye. The bacterium is able to survive on tomato volunteers and diseased plant debris. Seed may also serve as a medium for the survival and dissemination of the bacterium. Disease development is favored by temperatures of 24-30 degrees centigrade and by high precipitation. The bacterium is disseminated within fields by wind driven rain droplets, the clipping of transplants, and aerosols. It penetrates through stomata and wounds created by wind driven sand, insect punctures, or mechanical means.

Symptoms:

The bacterium affects all above ground parts. On the leaves, stems, and fruit spurs, the spots are generally brown and circular. The spots are water soaked during rainy periods or when dew is present. Lesions rarely to more than 3mm in diameter. On leaf lets the spots can easily be confused with early blight, gray leaf spot, or target spot. Bacterial spot lesions do not have concentric zones, as do target spots of early blight lesions, and they are generally darker in color and less uniformly distributed than gray spots lesions. Often a prominent halo surrounding pin point lesion is present with target spot and early blight but not with bacterial spot lesions. When conditions are optimal for disease development, spots on the leaves, petioles, and rachis coalesce to form long dark streaks. A general yellowing may occur on leaflets with lesions. Often the dead foliage remains on the plant, giving it a scorchy appearance. Fruit lesions begin as slightly raised blisters. As a spot increases in size it becomes brown, scab like, and slightly raised. Lesions may also be raised around the margins and sunken in the middle.

Control: Rotate fields in an attempt to avoid carry over on volunteer and crop residue.

Produce disease free transplants by raising them in areas where tomato and pepper have not been grown.

Use seed treatment (hot water at 50-55 degrees centigrade) to reduce possible transmission of the bacterium. Use copper fungicides as protectants to suppress bacteria.

Bacterial Speck: This is a bacterium that produces a number of compounds that help it infect and obtain nutrients from the tomato plant. One of these compounds is the plant-specific toxin coronatine, which is responsible for the yellow halo surrounding leaf lesions and the stunting of young seedlings. The major sources of infection for the bacteria are from seed and infected crop debris. Bacteria enter the plant through natural openings or wounds caused by wind-driven soil, insect or mechanical damage. Severe infections may cause defoliation. The bacterium may also be present on volunteer tomato plants and on contaminated equipment or surfaces (farm machinery, racks, greenhouse structures, tools).

Symptoms:

• Bacterial speck lesions may occur anywhere on the foliage, stems, or fruit. (Only green fruit is susceptible to infection, not red fruit)

• Fruit lesions are small, black, slightly raised and often surrounded by a narrow green-to-yellow halo

• Speck lesions on the fruit are usually superficial and can be scraped off.

• The bacteria are spread primarily by splashing water and wind-driven rain or mists produced during storms.

• In the field, spread by equipment or workers is probably of lesser importance than it is in the greenhouse, unless wounds are being opened up at the same time, as in pruning operation or when plants are injured by a cultivator.

Management/Control:

• Tomato seeds should be disinfected, using acid or chlorine treatment.

• Do not plant diseased transplants.

• Keep transplants from different seed lots and different transplant growers separate to avoid cross-contamination. Keep tomato transplants separated from other host crops such as peppers. Areas of potential contact include in the transplant greenhouse, during shipping or holding plants, and in the field.

• Clean and sanitize plant trailer (any equipment used for shipping or holding) between loads.

• Clean and sanitize the transplanter (surfaces that contact plants and trays) between fields and varieties.

• Transplanting crew cleans and sanitizes their hands at every break or changes to new disposable gloves.

• Clean and sanitize equipment that touches the crop between blocks of plants or between fields.

• In processing and unstaked fresh market tomatoes, eliminate hoeing and inter-row cultivating beyond 3 or 4 weeks after transplanting.

• When working with staked plants (pruning, tying), clean and sanitize tools between each plant.

• Crop scouts and other visitors instructed to clean and sanitize hands or wear gloves prior to entering each field. Wearing plastic booties which are changed after each field will also limit the spread of soil borne pathogens from field to field.

FUNGAL DISEASES

Alternaria spp.

Alternaria infection of seedlings can cause small stem lesions that can enlarge and eventually girdle the plant. Pythium and Phytophthora are water-molds. They are particularly destructive in wet and cool soil conditions. Pythium infection may result in pre- or post emergence damping-off, distinguished by dark-coloured, water-soaked lesions on roots or stem. Seed infected by Pythium becomes soft and water-soaked. Rhizoctonia solani persists in soil as a hard resting structure (sclerotia) and grows as microscopic threads through the soil. This damping off pathogen tends to prefer slightly warmer and dryer soil than the water molds. Often, Rhizoctonia will girdle the stems of susceptible crops slightly above and below the soil line. Lesions are tan to reddish-brown in colour.

Symptoms:

- Seeds infected prior to emergence rot and typically fail to produce a seedling.
- The seedlings do emerge; they are usually weak and lack vigour.
- Post-emergence infections cause the seedlings to rot at soil line.
- This usually occurs within 2-4 weeks of emergence (or transplanting).
- Affected plants tend to curl downward or melt into the soil.
- The plants also moldy seeds and lesions or cankers on the roots, hypocotyl or lower stem.

Control: • Ensure seedlings are grown in sterile soil-less mixture in the greenhouse.

- Do not over water seedlings.
- Before planting into the field, ensure all seedlings are healthy, disease-free and vigorous.

Disinfect the seed beds before sowing seeds.

Anthracnose: Anthracnose on tomatoes is caused by the fungus Colletotrichum coccoides, which is primarily a pathogen of the tomato fruit. As the fruit are ripening, the symptoms first become noticeable as small, circular indented areas, which later develop darkened centers. The diseased spots continue to grow larger with time as each infection site also spreads deeper into the fruit. With warm, moist and humid weather (from rainfall or over-head irrigation) the fungus produces salmon-colored spores that are exuded from the black fungal material in the center of the spots. These spores are spread by splashing water. Advanced infections occur on immature fruit and only later develop into visible lesions.

Control: • Follow a minimum 3-year rotation with non-solanaceous crops.

- Control weeds that can act as hosts.
- Use disease-free or treated seed.
- Properly timed fungicide sprays are effective at reducing losses to this disease.
- Fungicide sprays can help reduce disease, products containing chlorothalonil can be sprayed weekly to reduce infection.
- Follow label directions. There is a one day waiting period between spraying and picking.

EARLY BLIGHT

Symptoms: This particular disease is more serious on tomatoes during the hot weather. It occurs on all above ground parts and is destructive at all stages of crop development. Early blight begins as small brownish/ black lesions on leaves. The surrounding tissue becomes yellow covering the entire leaf resulting in partial defoliation. These spots typically develop first on the older leaves nearest the ground. Under favorable conditions for disease development, these diseases can cause extensive defoliation, resulting in sun-scalding of fruit and reduction in the numbers of fruit produced.

On stems, cankers form and occasionally fruits are infected causing premature fruit drop. Stem lesions on infected seedlings will continue to enlarge and subsequently girdle the seedlings. The most important diagnostic feature of early blight is the formation of dark, concentric rings within the lesion, giving the spots a target-like appearance. Often, several lesions coalesce, causing the leaf to turn yellow, dry up, and fall off the plant. Defoliation weakens the plant and exposes the fruit to sunscald injury.

Control:

• Sanitation measures. Field sanitation will reduce the amount of inoculum available for infection the following year. Deep-plough to bury tomato debris, or dead plants should be removed from the garden and destroyed.

• Crop rotation. Avoid planting tomatoes in the same area.

• Use clean planting. Clean seed and healthy transplants will help control the disease.

• Chemical control is the most effective method for control of late blight. This is due to the considerable distance over which the spores can travel. Spraying should start as soon as symptoms are observed, or preferably as soon as experience suggests that the conditions are right for disease development. During this time, protective fungicides e.g. Antracol (Propineb), Dithane

M-45 (Mancozeb) can be sprayed every 7-10 days. These can be alternated with systemic curative and protective fungicides like Ridomil (Metalaxyl) and Milrax (Propineb+Mancozeb) at 14 day intervals.

Late blight: This is one of the most destructive diseases of tomato. It is caused by a fungus Phytopthora infestans

Symptoms:

P. infestans attacks all above ground parts of tomato. Leaf lesions begin as water-soaked spots which rapidly enlarge to brown lesion engulfing the entire leaf which subsequently dries up. If the weather is cool, a gray/white moldy growth occurs on the underside of young lesions. Infected leaves shrivel, die and dry up. Infected areas on stems appear brown to black and entire vines may be killed in a short time when moist weather persists. Decaying stems emit a distinct foul rotting odor. Late blight can also develop on green tomato fruit, resulting in large, firm, brown, leathery-appearing lesions, often concentrated on the sides or upper fruit surfaces. If conditions remain moist, abundant white mold growth will develop on the lesions. and secondary soft-rot bacteria may follow, resulting in a slimy, wet rot of the entire fruit. Fruits also become infected in storage.

Control:

• Field sanitation to reduce the source of primary inoculum for adjacent tomato fields or plants.

• Avoid planting tomatoes after potatoes.

• Cultural methods of control include adequately deposing all tomato culls.

• Clean planting material. Use tomato transplants from a disease free nursery.

• Scout fields regularly to look for late blight. Scouting should be concentrated in low-lying areas, field edges along rivers or ponds, near the center of center-pivot irrigation rigs, in areas near trees or any area that is protected from wind where the leaves tend to remain wet longer.

• Chemical control is the most effective method for control of late blight.

This is due to the considerable distance over which the spores can travel.

***Spraying should start as soon as symptoms are observed, or preferably as soon as experience suggests that the conditions are right for disease development. During this time, protective fungicides e.g. Antracol (Propineb), Dithane M-45 (Mancozeb) can be sprayed every 7-10 days. These can be alternated with systemic curative and protective fungicides like Ridomil (Metalaxyl) and Milrax (Propineb+Mancozeb) at 14 day intervals.**

FUSARIUM WILT

Fusarium wilt is caused by Fusarium oxysporum f. sp. lycopersici. Fusarium is prevalent in fields where continuous cropping of tomato is practiced. Fusarium is
disseminated by seed and soil. It survives of weeds e.g. Amaranthus, Digitaria and Malva as reservoirs (Varela et al., 2003). It is more prevalent in warm 25 - 32 oC, acidic (pH 5.0-5.6) soils that are sandy and light in texture. This fungus can remain
in the soil indefinitely and several strains co-exist.

Symptoms: Seedlings infected by Fusarium wilt get stunted because the vascular system is affected. Typical symptoms include yellowing of leaflets on one side of the leaf while the other side remain healthy and green Subsequently, the lower leaves turn yellow and this progresses to other parts of the plant. These leaves eventually die and drop off. If the stem is cut vascular discoloration will be observed. Also, infected plants show vascular discoloration within the fruit (Jones, 1993). The vascular system of roots succumbs to the disease and decays.

Control:

- Cultural methods. Apply lime to increase the pH of the soil (pH 7.0).

Avoid excessive fertilization and cultivation. Use nitrate nitrogen rather than ammonium nitrogen.

- Crop rotation. Practice a 5 to 7 year rotation between successive crops Avoid rotating with peppers, eggplant, potatoes, sunflower, alfalfa, sweet clover.
- Field sanitation. Remove all debris and crop refuse after harvest and control susceptible weeds. Do not use flood water for irrigation?
- Chemical control. There is no chemical control for Fusarium wilt.
- Plant management. Stake plants to improve aeration so that they dry out soon and also to avoid contact between plants
- Resistant varieties. Most commercial cultivars have resistance to Fusarium wilt. Graft tomatoes onto resistant rootstocks.

VERTICILLIUM WILT

Verticillium wilt is caused by a soil borne fungus, Verticillium albo-atrum Reinke & Berhold. The pathogen has many hosts including cucurbits (cucumber, watermelon, muskmelon), pepper, potato, eggplant etc. is prevalent in cool temperatures (21-25 oC) and occurs in all tomato-growing regions.

Symptoms: plants show wilt symptoms during the warmest period of the day. plants show wilt symptoms during the warmest period of the day.

Control: Water the plant regularly, and when possible, provide afternoon shade. Fertilize on schedule, using a low-nitrogen, high-phosphorus fertilizer. Prune off dead and dying branches. You can often get rid of the verticillium wilt fungus in the soil by solarization.

SEPTORIA LEAF SPOT

Septoria leaf spot is caused by a fungus, Septoria lycopersici. It is one of the most destructive diseases of tomato foliage and is particularly severe in areas where wet, humid weather persists for extended periods. Septoria leaf spot usually appears on the lower leaves after the first fruit sets. It is one of the most destructive diseases of tomato foliage and is particularly severe in areas where wet, humid weather persists for extended periods. Septoria leaf spot usually appears on the lower leaves after the first fruit sets.

Control: Remove infected leaves immediately, and be sure to wash your hands thoroughly before working with uninfected plants. Fungicides containing either copper or potassium bicarbonate will help prevent the spreading of the disease. Begin spraying as soon as the first symptoms appear and follow the label directions for continued management.Use protective fungicides at regular intervals following manufacturers' recommendations.

POWDERY MILDEW

Powdery mildew of tomato, caused by the fungus Oidium lycopersicum. The spores are carried by wind to tomato plants. The disease usually is most severe late in the season. High relative humidity favors disease development.

Symptoms: It is characterized by white superficial mycelium on leaves and stems, yellowing, desiccation, necrosis and defoliation. Normally it begins with pale yellow spots on leaves. The spots soon become covered with white spores, which makes the leaves look like they have been dusted with flour. As this fungal disease advances, the whitish parts of the leaves turn brown and shrivel, becoming dry and brittle.

Control: Remove and destroy crop debris after harvest. Keep the field free of weeds and practice crop rotation. Irrigating regularly avoids drought stress in old crops. Use of products such Azoxystrobin, Myclobutanil, Thiophanate, Tebuconazole,Kresoxim-methyl and Sulfur based fungicides among others have given satisfactory control.

VIRAL DIESEASES

TOMATO SPOTTED WILT VIRUS

This disease that occurs commonly in the tropics. Plants infected with Tomato spotted wilt virus exhibit bronzing of the upper sides of young leaves, which later develop distinct, necrotic spots. Leaves may be cupped downward. Some tip dieback may occur. On ripe fruit chlorotic spots and blotches appear, often with concentric rings.

Control: Before farming use seeds resistant varieties. Resistant varieties generally do not require insecticide applications for thrips to control tomato spotted wilt. Use virus- and thrips-free transplants from greenhouses that manage thrips and inspect transplants.Manage thrips on transplants before planting. After transplanting, Avoid planting near crops infected with Tomato spotted wilt virus. Monitor for thrips and tomato spotted wilt symptoms. If thrips are present and symptoms are observed, manage thrips to minimize the spread of the virus within the field. Consider removing infected plants at the seedling stage. Control weeds in and around fields. After harvesting promptly remove and destroy old tomato plants and other host crops after harvest.Control weeds and volunteer plants in fallow fields, non-cropped, or idle land near next year's tomato field.

TOMATO YELLOW LEAF CURL VIRUS

Tomato leaf curl is a destructive viral disease caused by a virus in the Geminivirus family of plant viruses, tomato yellow leaf curl virus (TYLCV). It is spread by whiteflies and can be found in most areas where tomato is grown, causing significant losses.

Tomato is the major host of TYLCV, however, many other species are also infected, like capsicum, beans. Disease development is favoured by; High temperatures, low or no rainfall, presence of whiteflies Infected transplants, weedy fields.

Symptoms: severe stunting of young leaves and shoots, resulting in bushy growth of infected seedlings. Tomato plants leaves are curled, yellow (chlorotic) leaf margins, smaller leaves than normal. Plants that are infected early in their growth, may not form fruit.

Control: Removal of plants with initial symptoms may slow the spread of the disease. Infected plants should be immediately bagged to prevent the spread of the whiteflies feeding on those plants. Practice intercropping with other plants like onions, chives and garlic. Plant trap and repellant crops around your farm. Use sticky traps to control the victors population.

Keep weeds controlled within and around the garden site, as these may be alternate hosts for whiteflies. Spray low concentration sprays of a horticultural oil or canola oil will act as a whitefly repellent, reduce feeding and possibly transmission of the virus

TOMATO COMMON MOSAIC VIRUS

This viral disease is able to live on perennial weeds and is then spread by several insects including aphids, leafhoppers, whiteflies and cucumber beetles. Both cuttings and divisions from infected plants will be infected. The disease is spread into the plant via small wounds caused by mechanical injury, insect chewing, and grafting. Leftover plant debris is the most common way of spreading. Tomato mosaic virus of tomatoes can exist in the soil or plant debris for up to two years, and can be spread just by touch, a gardener who comes in contact with infected plant can carry the infection for the rest of the day. Always wash your hands with soap and disinfect tools after handling tomato plants to keep the disease from spreading.

Symptoms: Mottled light and dark green on leaves. Young tomato leaf that is mottled light and dark green, stunted growth, curled and malformed leaves. On a tomato seedling, they may appear yellow and stunted overall. Leaves may be curled, malformed, or reduced in size. Spots of dead leaf tissue may become apparent with certain cultivars at warm temperatures. Fruits may ripen unevenly. Reduced fruit number and size. Yellowish rings may form if fruit ripens in warm weather.

Control: Use certified disease, Disinfect tools regularly, Disinfect tools regularly, Tobacco in cigarettes and other tobacco products may be infected with either ToMV or TMV, both of which could spread to the tomato plants.

NUTRITIONAL DISORDERS

Nutrient disorders result from chemical imbalances in plants caused by prolonged insufficient or excessive supply of chemical elements. Nutrient deficiency symptoms develop when an essential element for plant growth is below the optimum concentration required for growth and development.

Nitrogen: N (nitrogen) deficiency of tomato crop is typically characterizes by older leaves that gradually change from green to yellowish or paler green. These leaves will later become yellow, and under extreme nitrogen deficiency they are likely to become bright white-yellow.

Phosphorous: is most often manifested as purpling of the leaves, particularly the leaf veins. In severe cases the whole plant may take on a purple hue. Tomato roots growing in cold soil, either in the greenhouse or the field, take up phosphorus poorly. Deficient plants lose vigor and yield poorly.

Potassium: is generally characterized by a marginal chlorosis progressing into a dry leathery tan scorch on recently matured leaves, even if potassium is given to the plants. Because potassium is very mobile within the plant, symptoms only develop on young leaves in the case of extreme deficiency.

Calcium: A lack of calcium shows up as young leaves curling inwards and lacking colour, and is often a problem in acid soils. 'Blossom end rot' in tomatoes is caused by this condition.

Magnesium: deficiency start on older leaves. They show interveinal chlorosis on the leaf margins and some whitish to light brown necrotic dots. If deficiency is severe, interveinal chlorosis progresses from the margins to the middle of the leaflets.

Iron: iron-deficient leaves show strong chlorosis at the base of the leaves with some green netting. The most common symptom for iron deficiency starts out as an interveinal chlorosis of the youngest leaves, evolves into an overall chlorosis, and ends as a totally bleached leaf.

NUTRITIONAL DISORDERS

Magnesium

Iron

Potassium

Nitrogen

Calcium

Phosphorus

HARVESTING

The end-use of the produce and the distance to the market will determine when to start harvesting. For example, tomatoes for processing are harvested when they are fully mature, those to be shipped long distance is harvested at a less mature stage, while the crop for local markets is picked at a more mature stage. The maturity days vary depending on the variety in question.

The harvested fruits are stored into holding containers normally plastic or wooden crates placed under shade in the farm.

OUR DISTRIBUTION NETWORK

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